Chenjia Bai（白辰甲）

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Research Topic: **Reinforcement Learning** (RL): Offline RL, Efficient Exploration, Representation Learning, Multi-Goal RL, Robotics.

Education

Mar 2021 – Mar 2022	**Joint Ph.D. Training,** Computer Science, **University of Toronto**, advised by Prof. Animesh Garg, funded by China Scholarship Council.
Sep 2017 - July 2022	**Ph.D. Candidate**, School of Computer Science and Technology, HIT, advised by Prof. Peng Liu.
Sep 2015 - July 2017	**Master**, School of Computer Science and Technology, HIT Major: Computer vision, Data mining
Sep 2011 - July 2015	**Bachelor**, School of Computer Science and Technology, HIT

Publication

[1] **Chenjia Bai**, Lingxiao Wang, Zhuoran Yang et al Pessimistic Bootstrapping for Uncertainty-Driven Offline Reinforcement Learning. *International Conference on Learning Representations (**ICLR**). 2022.* (OpenReview link, ranking Top 8% scores, with an accepted rate of 25%)

[2] **Chenjia Bai**, Lingxiao Wang, Lei Han, et al. Dynamic Bottleneck for Robust Self-Supervised Exploration. *Neural Information Processing Systems (**NeurIPS**)*, 2021 (CCF-A)

[3] **Chenjia Bai,** Lingxiao Wang, Lei Han, et al. Principled Exploration via Optimistic Bootstrapping and Backward Induction. *International Conference on Machine Learning (**ICML**)*, 2021 (CCF-A)

[4] **Chenjia Bai**, Peng Liu, Kaiyu Liu, et al. Variational Dynamic for Self-Supervised Exploration in Deep Reinforcement Learning. *IEEE Transactions on Neural Networks and Learning Systems.(**TNNLS**)*. 2021. (CCF-B, JCR-1, IF=10.5)

[5] **Chenjia Bai,** Lingxiao Wang, Yixin Wang, et al. Addressing Hindsight Bias in Multi-Goal Reinforcement Learning. *IEEE Transactions on Cybernetics (**TCYB**)*, 2021. (CCF-B, JCR-1, IF=11.4)

[6] Peng Liu, **Chenjia Bai**, Wei Zhao, et al. Generating attentive goals for prioritized hindsight reinforcement learning. *Knowledge-based Systems.* 2020 (CCF-C, JCR-1, IF=8.0, 导师一作)

[7] **Chenjia Bai**, Peng Liu, Wei Zhao, et al. Guided goal generation for hindsight multi-goal reinforcement learning. *Neurocomputing*. 2019, 359: 353-367. (CCF-C, JCR-1, IF=5.7)

[8] **Chenjia Bai**, Peng Liu, Wei Zhao, et al. 基于 TD-error 自适应校正的深度 Q 学习主动采样方法. *计算机研究与发展*. 2019 (一级学报)

Under Review

[9] **Chenjia Bai**. et al. Monotonic Quantile Network for Worst-Case Offline Reinforcement Learning. *IEEE Transactions on Neural Networks and Learning Systems.(**TNNLS**)*. 2021

[10] Tianpei Yang*, Hongyao Tang*, **Chenjia Bai***, et al. Exploration in Deep Reinforcement Learning: A Comprehensive Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence (**T-PAMI**)* (*Equal Contribution, arxiv link, 机器之心)

Research Experience

Offline Reinforcement Learning

- Proposing a purely uncertainty based offline RL method without policy constraints.
- Proposing a risk-averse policy learner with conservative non-crossing quantile regression in offline RL.

Exploration & Representation Learning for Reinforcement Learning

- Proposing robust exploration based on information bottleneck, has theoretical guarantee under the approximation of linear value function, performs robust in high-dimensional tasks with noises.
- Proposing an exploration method based on ensemble Q-networks, obtains SOTA results in Atari games.
- Proposing variational model to learn the stochasticity and multi-modality of dynamics. Exploring the environment through information gain in embedding spaces. Conduct experiments in real-world robotics.

Multi-Goal Reinforcement Learning

- Analyzing the hindsight bias in multi-goal RL and proposing a bias-correction method for robot learning.
- Proposing to use transformer and attention mechanism to generate goals for multi-goal RL in robotics.
- Proposing a probabilistic generative model to learn the relationship between trajectory and return.

Internship

Oct 2020 - May 2021	Tencent Robotics X Research Intern, advised by Dr. Lei Han.	Tencent, Shenzhen
July 2020 - Sep 2020	Huawei Noah's Ark Lab Research Intern, advised by Prof. Jianye Hao.	Huawei, Beijing

Project Experience

- National Defense Science and Technology of China, “Brain like Navigation and Target Recognition Technology”.
  - Leader: Peng Liu Funding: 2.80 million CNY
  - Conducted research on Navigation and Decision based on Reinforcement Learning.
- Key Program of the Natural Science Foundation of China, “Research on Key Technologies of Mars Spacecraft and Unknown Exploration”.
  - Leader: Dewei Tang Funding: 3.20 million CNY
  - Conducted research on Reinforcement Learning for Path Planning in unknown areas.

Academic Services

- Reviewer of conferences including *NeurIPS, ICML, ICLR, AAAI.*
- Teaching assistant of High-level language programming.